

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I KURUBA PARTHASARATHI (192312344) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- Compute n and $\phi(n)$
- Choose $e=7$ and find the private key d
- Encrypt the message $M=9$
- Decrypt the ciphertext and verify the result
- Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

- Encrypt using ECB and CBC modes (Assume IV = 0000)

- b. Compare the ciphertext outputs
- c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison

Given:

- a. RSA key size = 1024 bits
- b. ECC key size = 160 bits (equivalent security)
- c. Device processing capacity = limited (IoT device)
- d. Analyze computational cost for both systems
- e. Estimate which consumes less power and memory
- f. Justify which cryptosystem is better for the given scenario with reasoning

ANSWEERS

1. RSA Encryption Analysis

Given:

$$p = 11, q = 13$$

a) Compute n and $\phi(n)$

$$n = p \times q = 11 \times 13 = 143$$

$$\phi(n) = (11 - 1)(13 - 1) = 10 \times 12 = 120$$

Answer:

$$n = 143$$

$$\phi(n) = 120$$

b) Choose $e = 7$ and find private key d

Find d such that:

$$7d \equiv 1 \pmod{120}$$

$$7 \times 103 = 721 = 120 \times 6 + 1$$

Therefore,

Private key: $d = 103$

c) Encrypt message $M = 9$

Formula:

$$C = M^e \pmod{n}$$

$$C = 9^7 \pmod{143} = 48$$

Ciphertext = 48

d) Decrypt ciphertext

Formula:

$$M = C^d \pmod{n}$$

$$M = 48^{103} \pmod{143} = 9$$

Recovered message = 9

e) Effect of changing e

- Small e gives faster encryption.
- e must satisfy $\gcd(e, \phi(n)) = 1$.
- Very small values may introduce attacks if padding is absent.
- Common secure choice: 65537.

2. Block Cipher Mode Evaluation

Given

Plaintext Blocks

$$P_1 = 1010$$

$$P_2 = 1010$$

$$P_3 = 1111$$

$$P_4 = 1010$$

$$\text{Key } K = 1100$$

IV = 0000

Assume encryption = XOR with key.

(a) ECB Mode

Formula

$$C = P \oplus K$$

Block 1

1010 XOR 1100 = **0110**

Block 2

1010 XOR 1100 = **0110**

Block 3

1111 XOR 1100 = **0011**

Block 4

1010 XOR 1100 = **0110**

ECB Ciphertext

0110 0110 0011 0110

CBC Mode

Block 1

1010 XOR 0000 = 1010

1010 XOR 1100 = 0110

$C_1 = 0110$

Block 2

1010 XOR 0110 = 1100

1100 XOR 1100 = 0000

$C_2 = 0000$

Block 3

1111 XOR 0000 = 1111

1111 XOR 1100 = 0011

$C_3 = 0011$

Block 4

1010 XOR 0011 = 1001

1001 XOR 1100 = 0101

$C_4 = 0101$

CBC Ciphertext

0110 0000 0011 0101

b) Compare Ciphertexts

ECB:

0110 0110 0011 0110

CBC:

0110 0000 0011 0101

c) Which mode hides patterns?

CBC hides repeated plaintext patterns because every block depends on the previous ciphertext and IV, making it more secure than ECB.

3. DES vs 3-DES vs Blowfish

Given

DES:

Key = 56 bits

Time = 2 ms/block

3DES:

Key = 168 bits

Time = 6 ms/block

Blowfish:

Key = 128 bits

Time = 1 ms/block

For **100 blocks**

a) Total Encryption Time

DES

$100 \times 2 = 200 \text{ ms}$

3DES

$100 \times 6 = 600 \text{ ms}$

Blowfish

$100 \times 1 = 100 \text{ ms}$

b) Trade-offs

Algorithm
m

Security

Speed

DES

Low

Medium

3DES

High

Slow

Blowfish

High

Fast

c) Best Algorithm

Blowfish

Reason:

- Fastest
- Strong security
- Suitable for secure real-time communication

4. Diffie-Hellman Key Exchange

Given

$$p = 23$$

$$g = 5$$

User A private key = 6

User B private key = 15

a) Public Key of A

$$A = 5^6 \cdot 23$$

$$15625 \cdot 23 = 8$$

$$A = 8$$

b) Public Key of B

$$B = 5^{15} \cdot 23 = 19$$

$$B = 19$$

c) Shared Secret

For A

$$K = 19^6 23 = 2$$

For B

$$K = 8^{15} 23 = 2$$

Shared Secret = 2

d) Man-in-the-Middle Attack

An attacker intercepts the exchanged public keys and replaces them with their own public keys. Both users unknowingly establish separate secret keys with the attacker, allowing the attacker to read or modify messages.

e) Prevention

- Digital Signatures
- Digital Certificates (PKI)
- Authenticated Diffie-Hellman
- TLS/SSL

5. RSA vs ECC Comparison

Given

RSA

Key size = 1024 bits

ECC

Key size = 160 bits

Device

Limited IoT processing power

a) Computational Cost

- RSA requires larger keys and more computations.
- ECC performs equivalent security with much smaller keys, making computations faster.

b) Power Consumption

- RSA consumes more CPU time, memory, and battery.
- ECC requires less computation and therefore uses less power and memory.

c) Better Cryptosystem

ECC is the better choice for the IoT device.

Reason:

- Smaller key size (160 bits)
- Equivalent security to RSA 1024-bit
- Faster computation
- Lower memory usage
- Lower power consumption
- Ideal for embedded and IoT devices

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

strong justification

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation